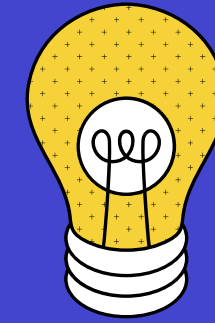
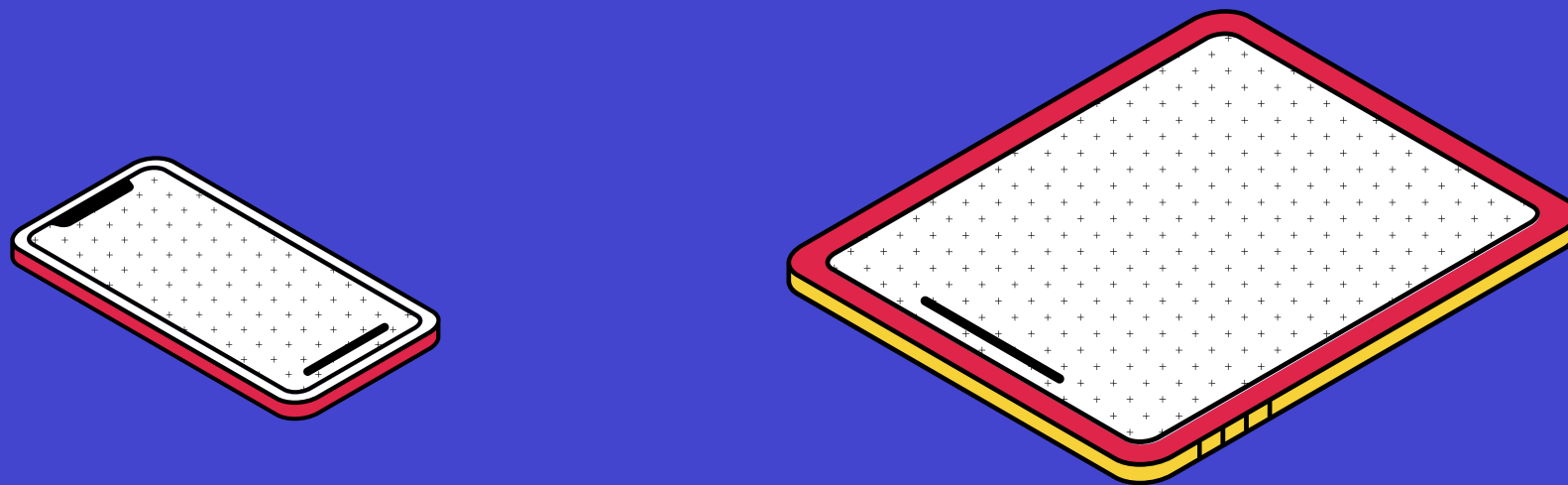


# ONLINE SHOPPING SYSTEM



Design and Implementation using the  
Waterfall Methodology



**Course:** Software Development Methodologies

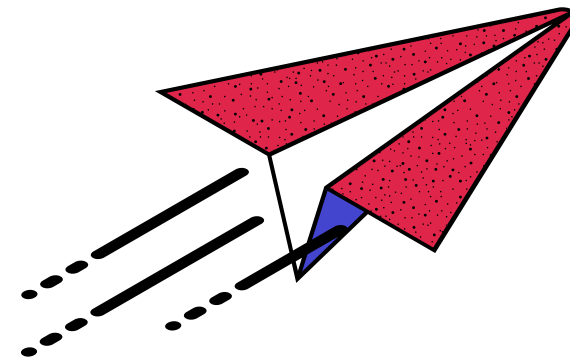
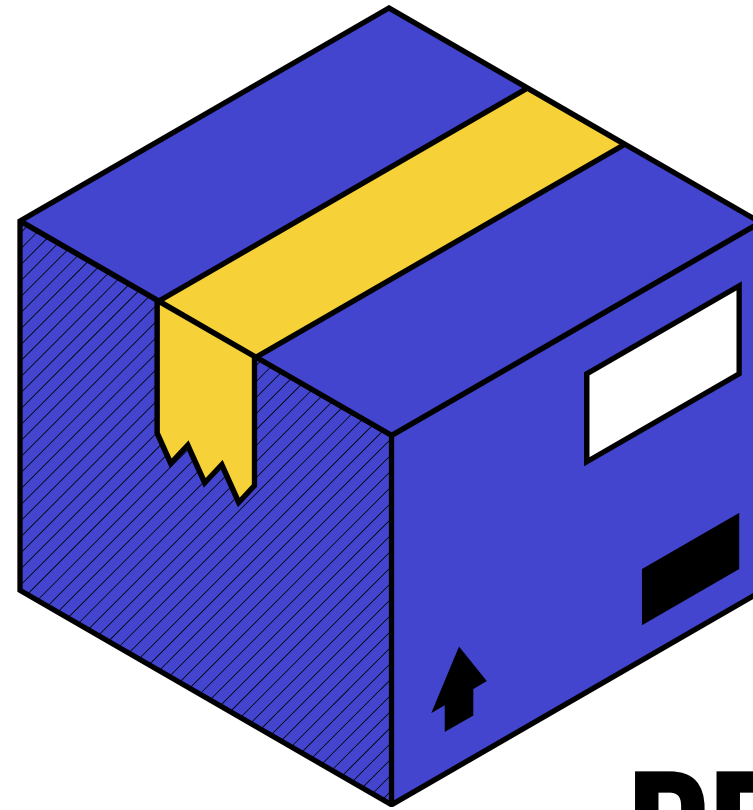
**Authors:** Dilnaz, Zhanara, Otabek, Hicham, Elif

**Supervisor:** Marcin Kacprowicz

# PROJECT OVERVIEW

**The Online Shopping System is a web-based application that allows users to browse, search, and purchase products online.**

**It supports both customer functionalities and administrative management, and was developed as a functional prototype following the Waterfall development methodology.**



## PROJECT OBJECTIVE

- **Allows users to register, log in, and place orders**
- **Provides product browsing and cart management**
- **Includes admin features for managing products and orders**
- **Follows all stages of the Waterfall Model**
- **Demonstrates structured software development and documentation**

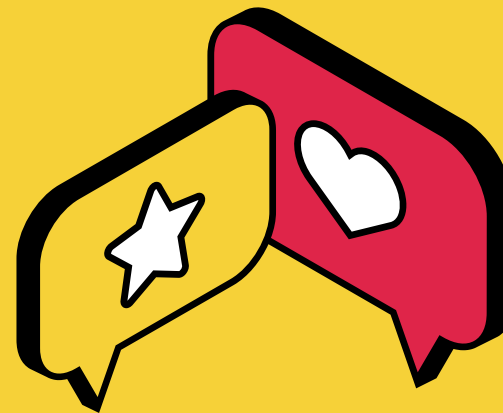
# PROJECT SCOPE

## In-Scope Features

- User registration and authentication
- Product browsing and search
- Shopping cart management
- Order placement and checkout
- Admin product and order

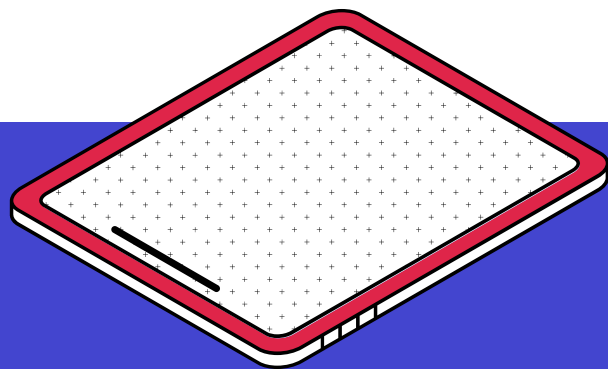
## Out-of-Scope Features

- Real payment gateway integration (PayPal, Stripe)
- Delivery tracking systems
- Mobile application development



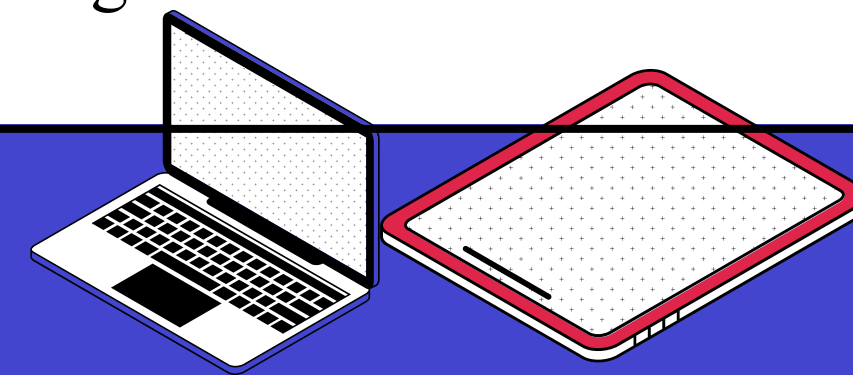
# Functional Requirements

| ID  | Requirements            | Description                                                              |
|-----|-------------------------|--------------------------------------------------------------------------|
| FR1 | User Registration/Login | The system allows users to register an account and log in securely.      |
| FR2 | Product Catalog         | Displays available products with details such as name, image, and price. |
| FR3 | Add to Cart             | Users can add items to their cart and view the selected products.        |
| FR4 | Checkout Process        | Users can proceed to checkout and confirm their order.                   |
| FR5 | Admin Management        | Administrators can manage products and review customer orders.           |



# Non - functional Requirements

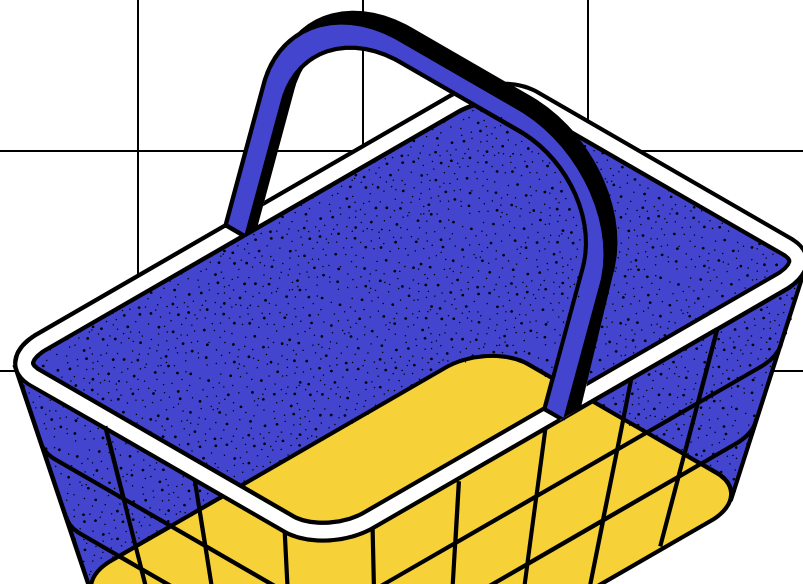
| ID   | Requirements  | Description                                                                                                    |
|------|---------------|----------------------------------------------------------------------------------------------------------------|
| NFR1 | Usability     | The interface should be user-friendly and easy to navigate                                                     |
| NFR2 | Performance   | The system should respond to user requests time $\leq 2$ seconds.                                              |
| NFR3 | Security      | User credentials and data must be stored securely                                                              |
| NFR4 | Scalability   | The system architecture should allow future feature expansion                                                  |
| NFR5 | Compatibility | The system should be compatible with major browsers, operating systems, and devices (desktop, tablet, mobile). |
| NFR6 | Reliability   | The system should handle errors without crashing or losing data                                                |



# **WATERFALL MODEL**

**The Waterfall Model was chosen because:**

- **Requirements are clear and stable**
- **The project is small and academic**
- **Each phase is well documented**
- **Progress is easy to track**



## **WATERFALL STAGES**

- 1. Requirement Analysis – Gathered functional and non-functional requirements**
- 2. System Design – Designed architecture, UI, and database structure**
- 3. Implementation – Developed system modules sequentially**
- 4. Testing – Verified functionality and usability**
- 5. Deployment & Maintenance – Final review and minor improvements**

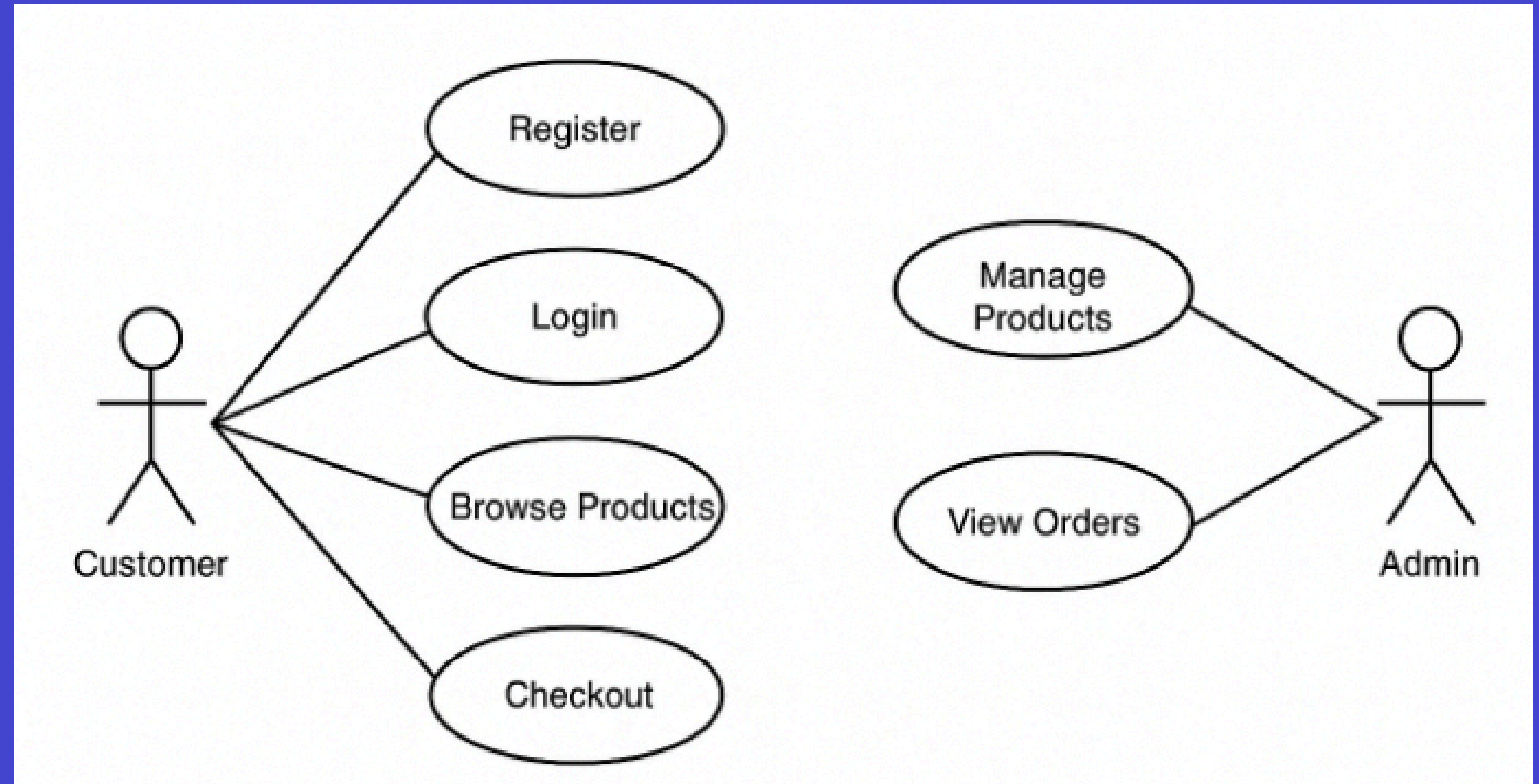
# USE CASE DIAGRAM

**Customer can:**

- Register
- Login
- Browse Products
- Checkout

**Admin can:**

- Manage Products
- View Orders



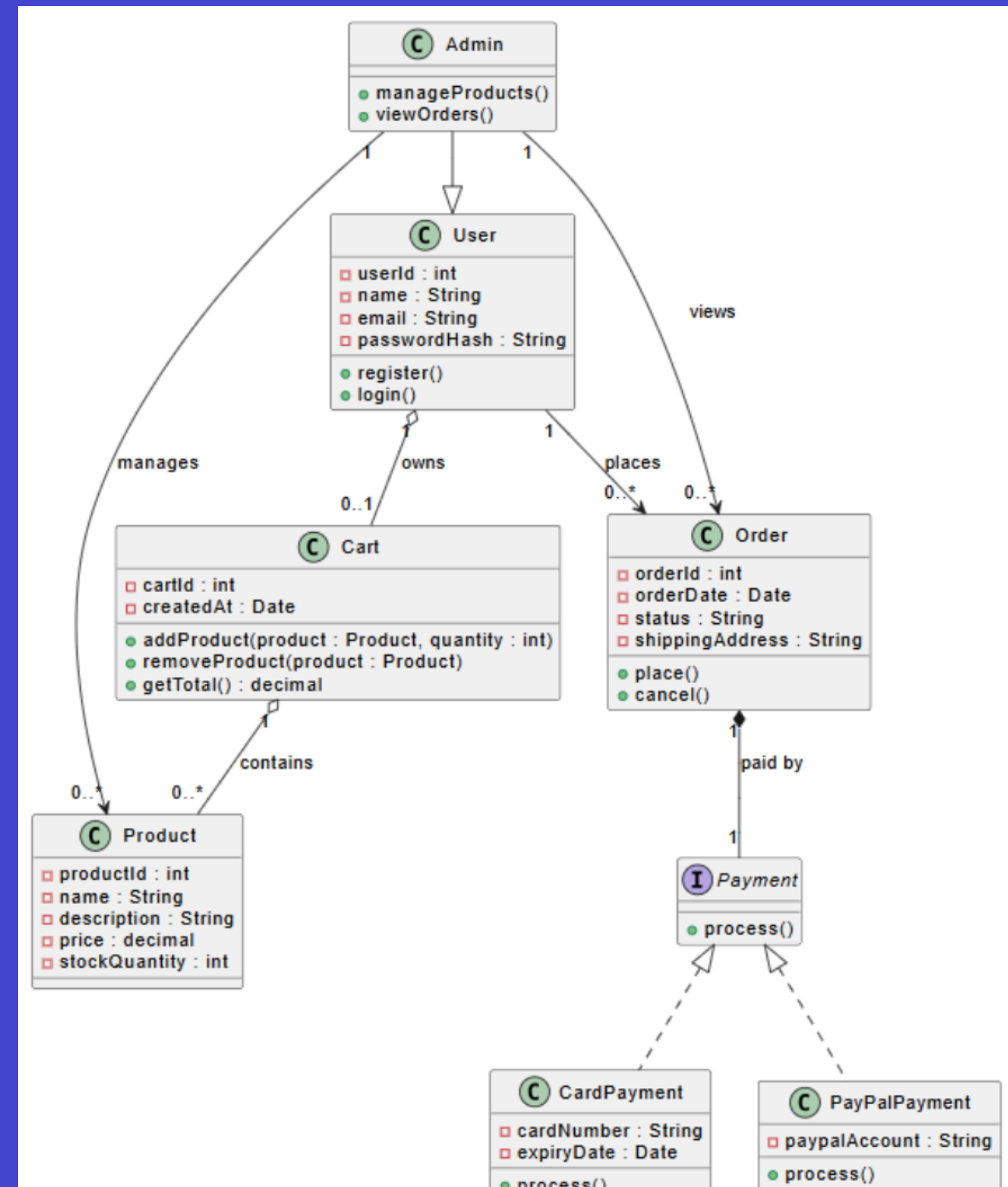
# CLASS DIAGRAM

Main system classes include:

- User – Customer information and authentication
- Admin – Special user with management permissions
- Product – Product details and stock
- Cart – Selected items and total calculation
- Order – Completed purchase details
- Payment – Abstract payment processing

Class Relationships

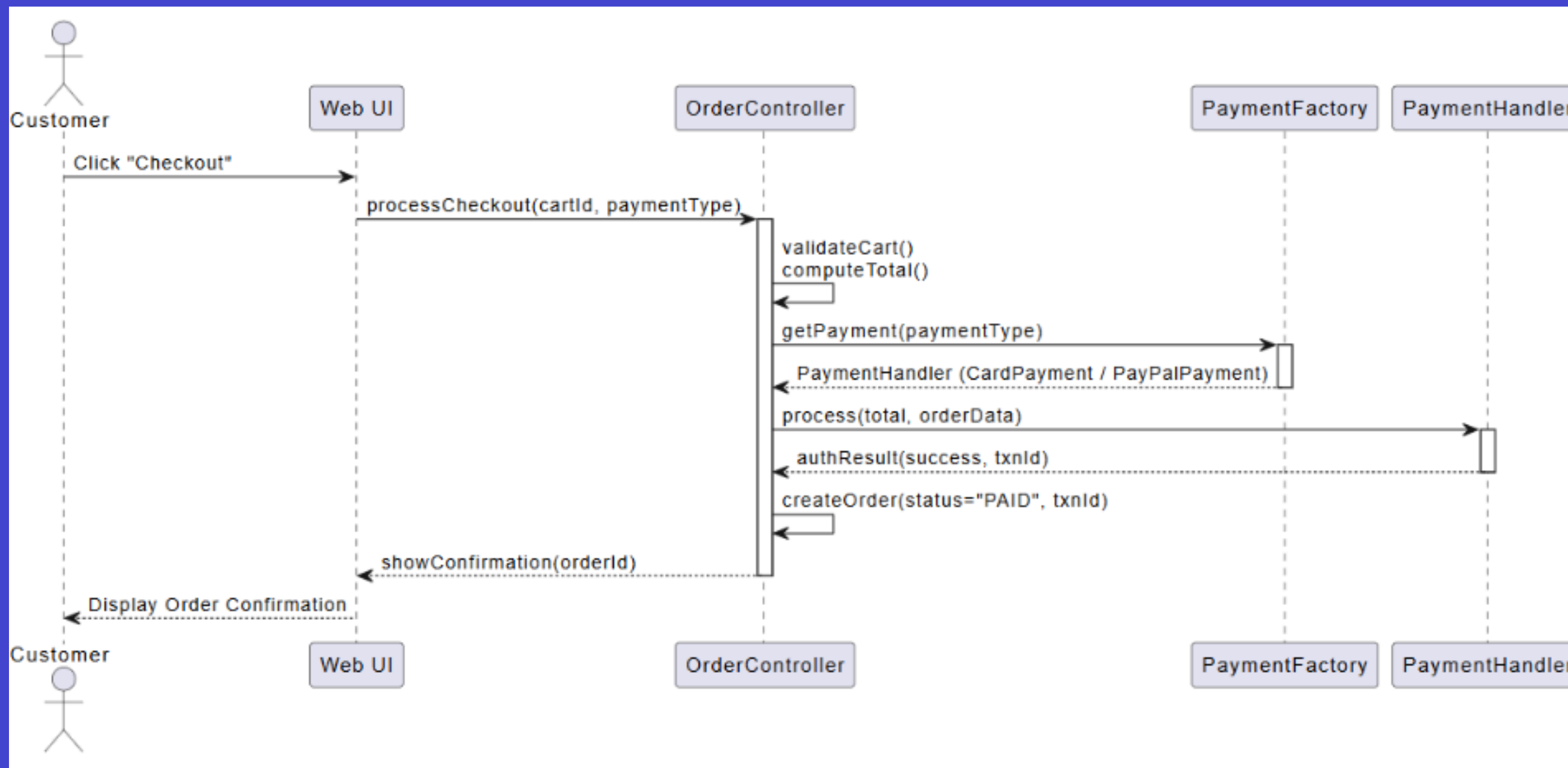
- A user has one active cart
- A cart contains multiple products
- A user can place multiple orders
- Each order has exactly one payment method
- Admin inherits from User
- Admin manages products and views orders



# SEQUENCE DIAGRAM

The sequence diagram shows the checkout process, including:

- Customer initiating checkout
- Web UI sending request
- OrderController processing logic
- PaymentFactory selecting payment method
- Payment execution and confirmation



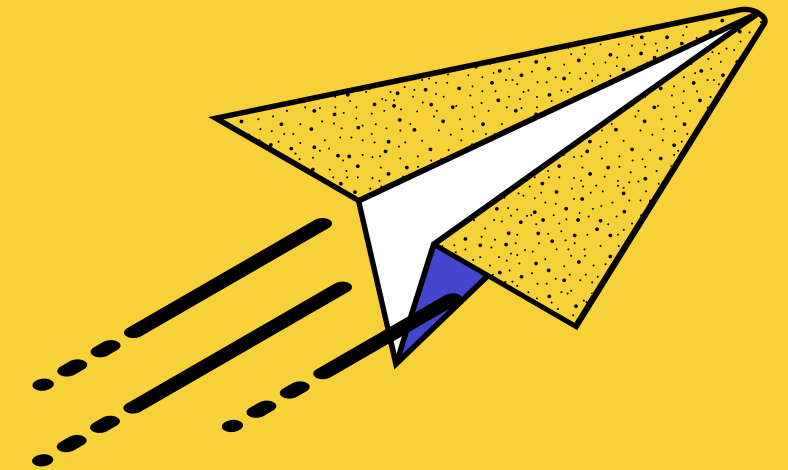
# ARCHITECTURE DIAGRAM

## MVC ARCHITECTURE

The system follows the Model–View–Controller (MVC) pattern:

- **Model:** Handles data and business logic
- **View:** Displays pages and user interfaces
- **Controller:** Processes user requests and system flow

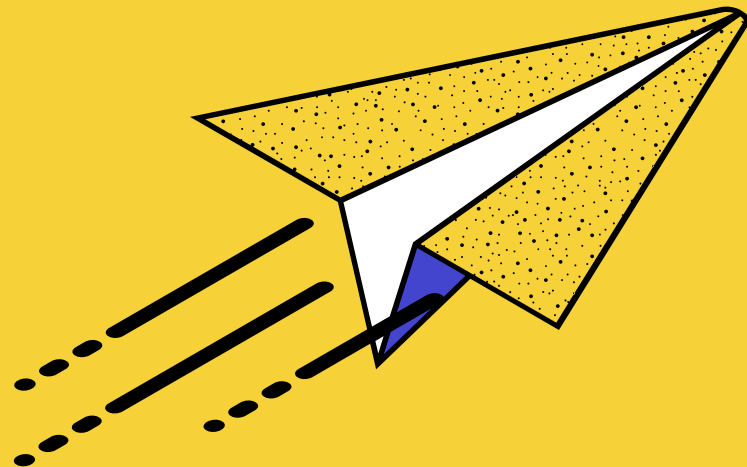
This separation improves maintainability and scalability.



# MVC FLOW OF DATA

1. **User interacts with the View**
2. **Controller receives the request**
3. **Model processes data and logic**
4. **Controller returns results to the View**

## WHY MVC WAS USED

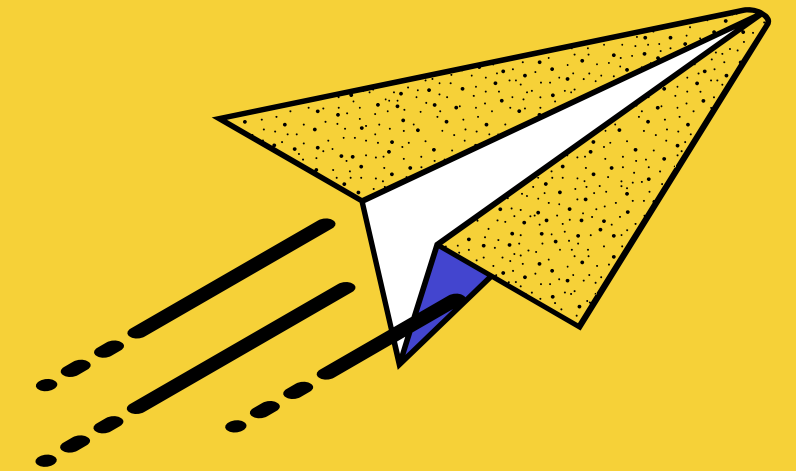


- **Clear separation of concerns**
- **Easier maintenance and extension**
- **Supports teamwork**
- **Industry-standard architecture**

# FACTORY DESIGN PATTERN

## The Factory Pattern is used to:

- Centralize object creation
- Reduce coupling between components
- Improve system flexibility
- Simplify future extensions



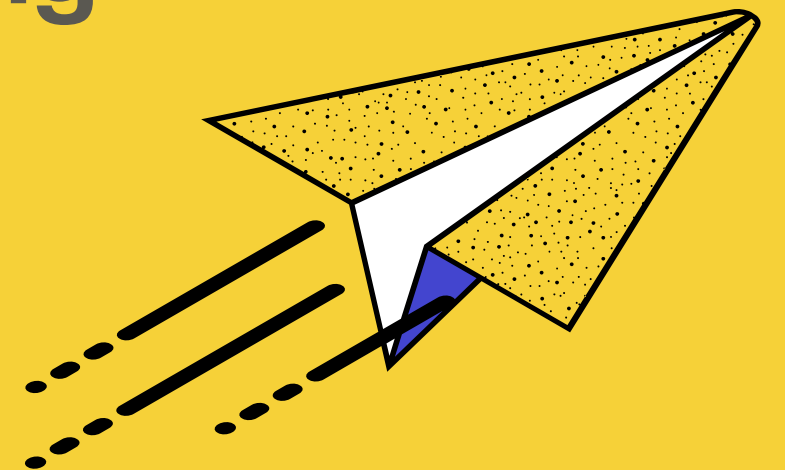
## Purpose of the Factory Pattern

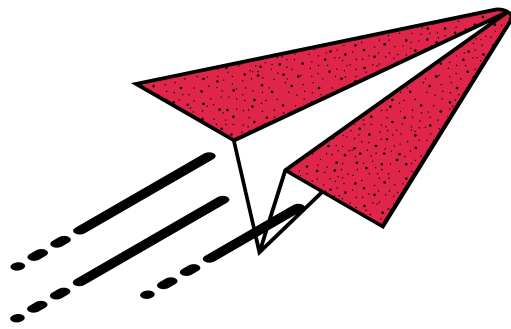
- Encapsulates object creation logic in one place
- Reduces coupling between components
- Makes the system easier to extend with new object types
- Improves readability and maintainability

# FACTORY DESIGN PATTERN

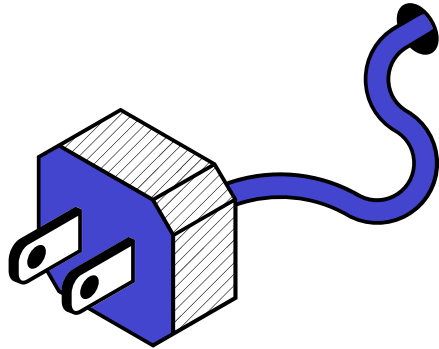
## How the Factory Pattern is used in the system?

In parts of the system where different processing classes may be needed (for example: different types of order handlers, or different types of product processors), the Factory returns the appropriate object without exposing creation details.

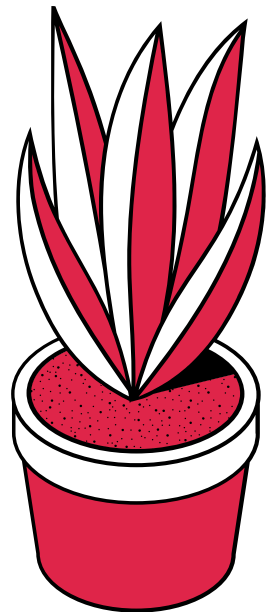


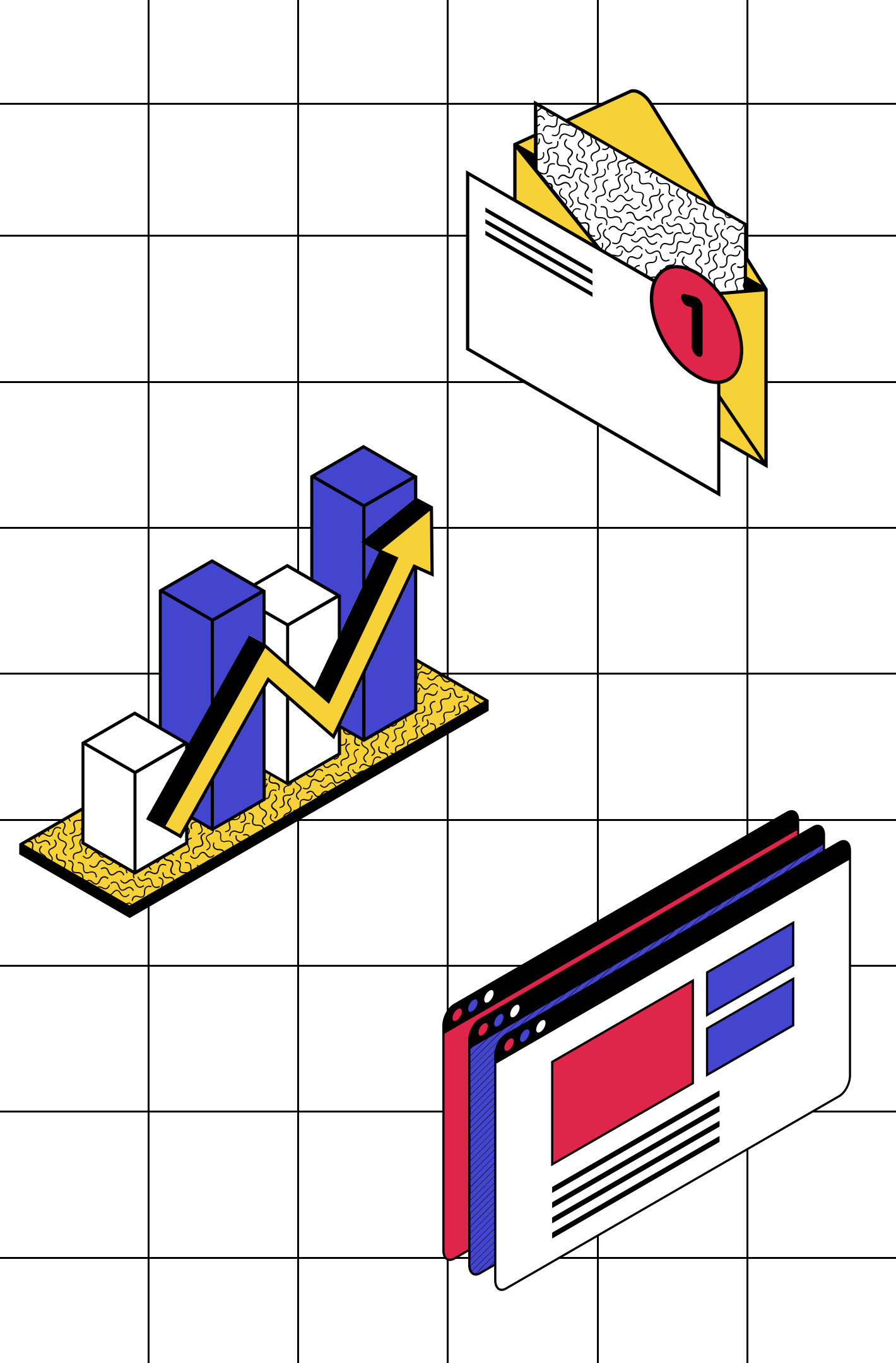


# TESTING STRATEGY AND RESULTS



| Testing Methodology                                                             | Test Types                                                                                                                                             | Testing Tools                           |
|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| Testing was performed after implementation, following the Waterfall methodology | <div>Unit Testing</div> <ul style="list-style-type: none"><li>• User login and authentication</li><li>• Product browsing functionality</li></ul>       | Manual testing for user flows           |
| Focused on validating functional requirements and system usability              | <div>Functional Testing</div> <ul style="list-style-type: none"><li>• Add to cart operations</li><li>• Checkout and order placement workflow</li></ul> | JUnit (for basic unit tests)            |
| Ensured all core shopping features work as expected                             | <div>Usability Testing</div> <ul style="list-style-type: none"><li>• Ease of navigation</li><li>• Clarity of user interface</li></ul>                  | Browser-based testing for UI validation |





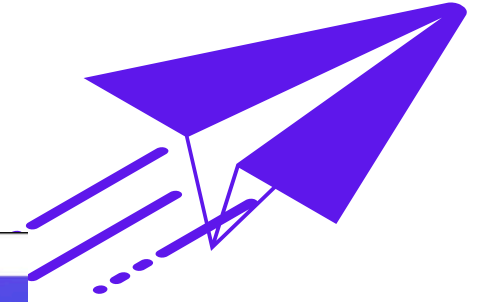
# TEST SCENARIOS & RESULTS

| TEST ID | SCENARIO                      | RESULT |
|---------|-------------------------------|--------|
| T1      | User registration & login     | Passed |
| T2      | Product browsing & search     | Passed |
| T3      | Add to cart functionality     | Passed |
| T4      | Checkout & order confirmation | Passed |

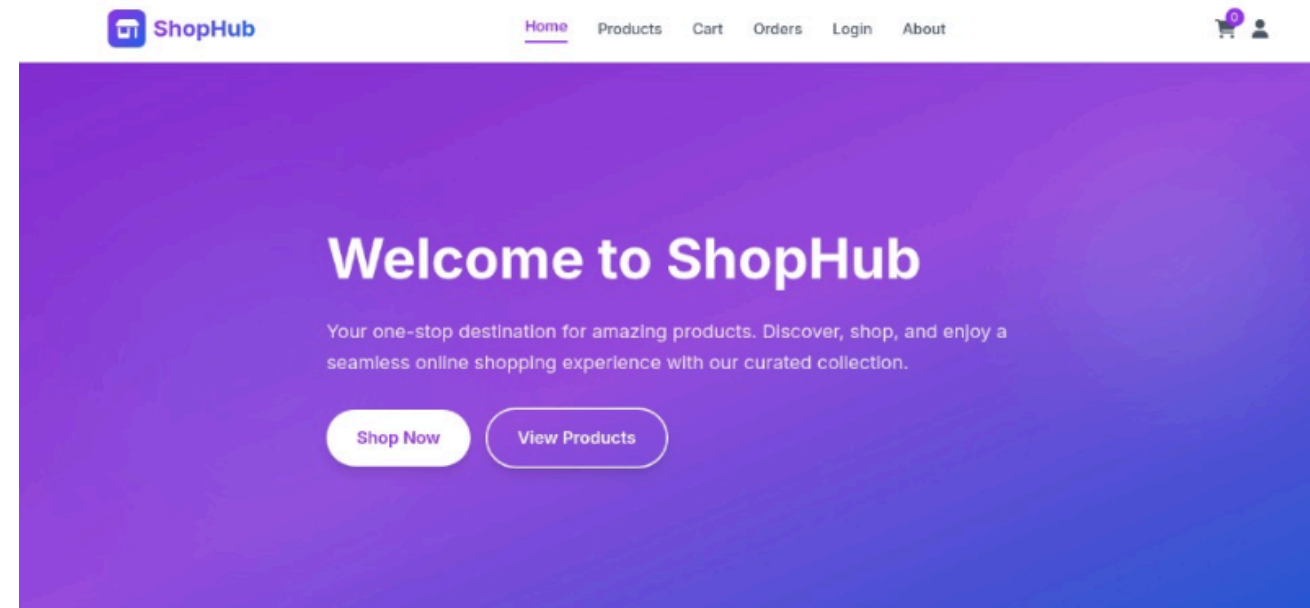
## Testing Outcome

- System meets all functional requirements
- No critical defects detected
- Application is stable and ready for deployment

# USER EXPERIENCE: FROM BROWSE TO PURCHASE



The sign-in form features a blue circular icon with a white person silhouette at the top. Below it, the text 'Sign In' is displayed in bold. A subtitle reads 'Enter your credentials to access your account'. The form includes two input fields: 'Email Address' with the placeholder 'john.doe@example.com' and 'Password' with the placeholder 'Enter your password'. A 'Remember me' checkbox and a 'Forgot password?' link are positioned below the password field. A large blue button with a white right-pointing arrow and the text 'Sign In' is at the bottom. Below this, a separator line reads 'Or continue with'. Two buttons for social login are provided: 'Continue with Google' and 'Continue with Facebook'. At the very bottom, a link says 'Don't have an account? Create one here'.

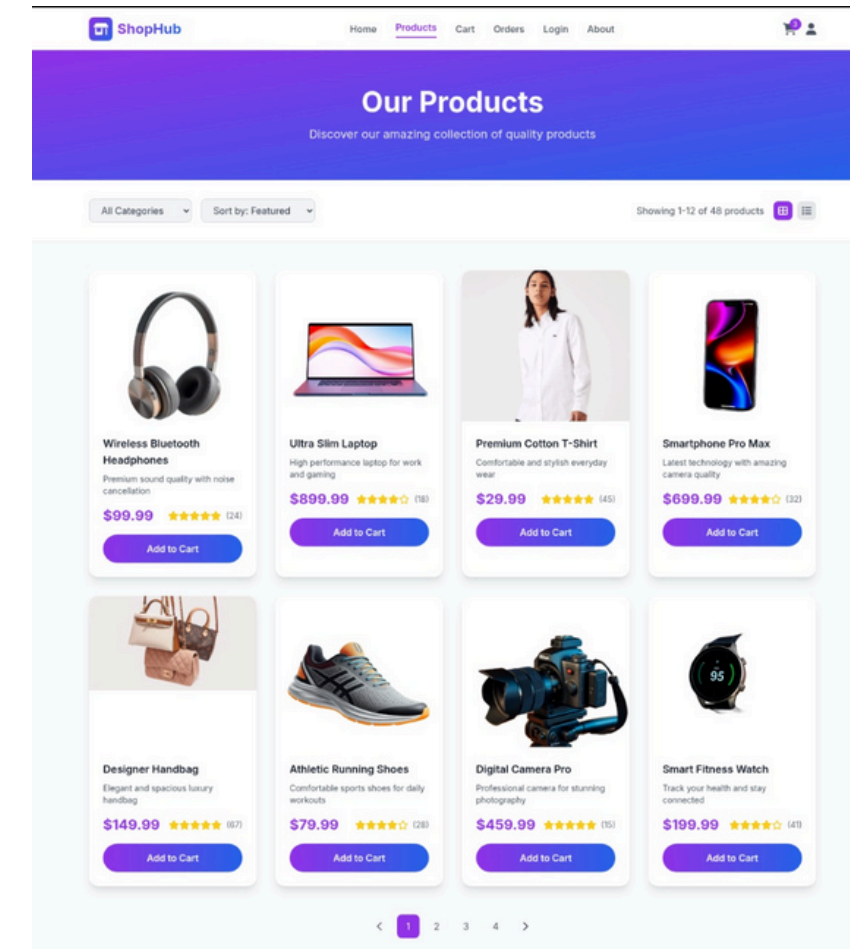


## Home & About

- Home page introduces the platform and key features
- About page explains project purpose and technologies used

## Authentication

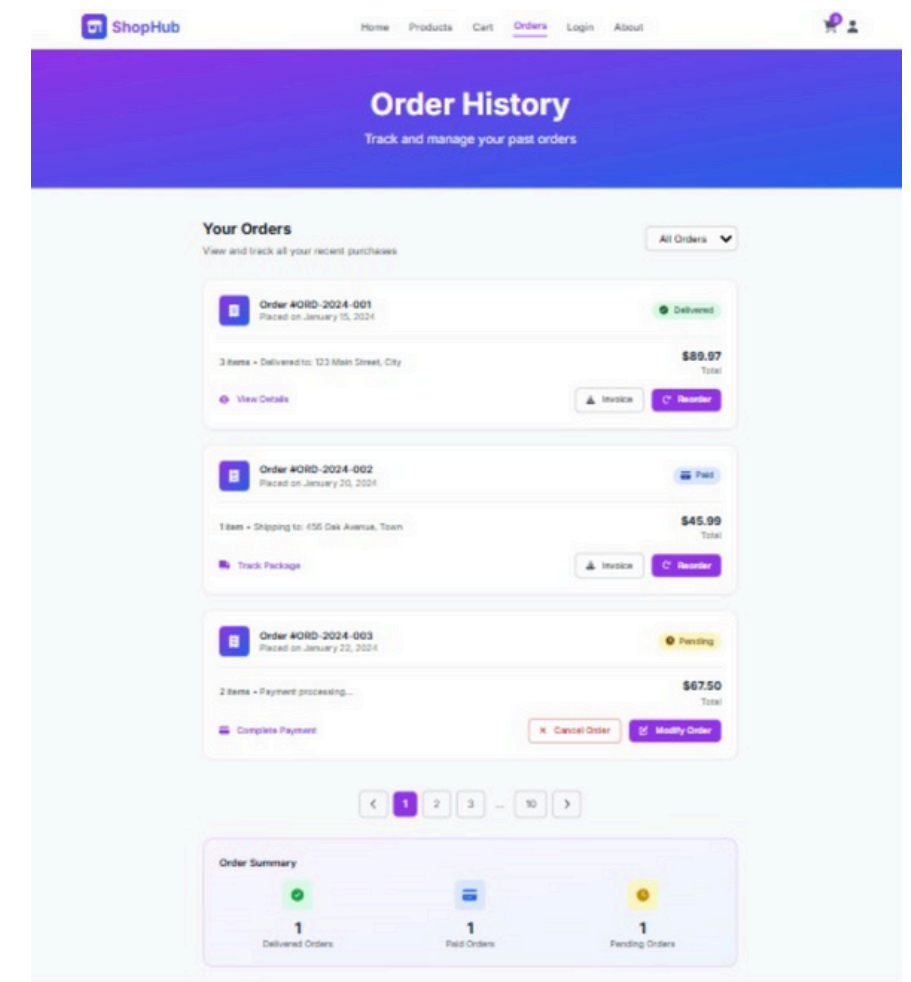
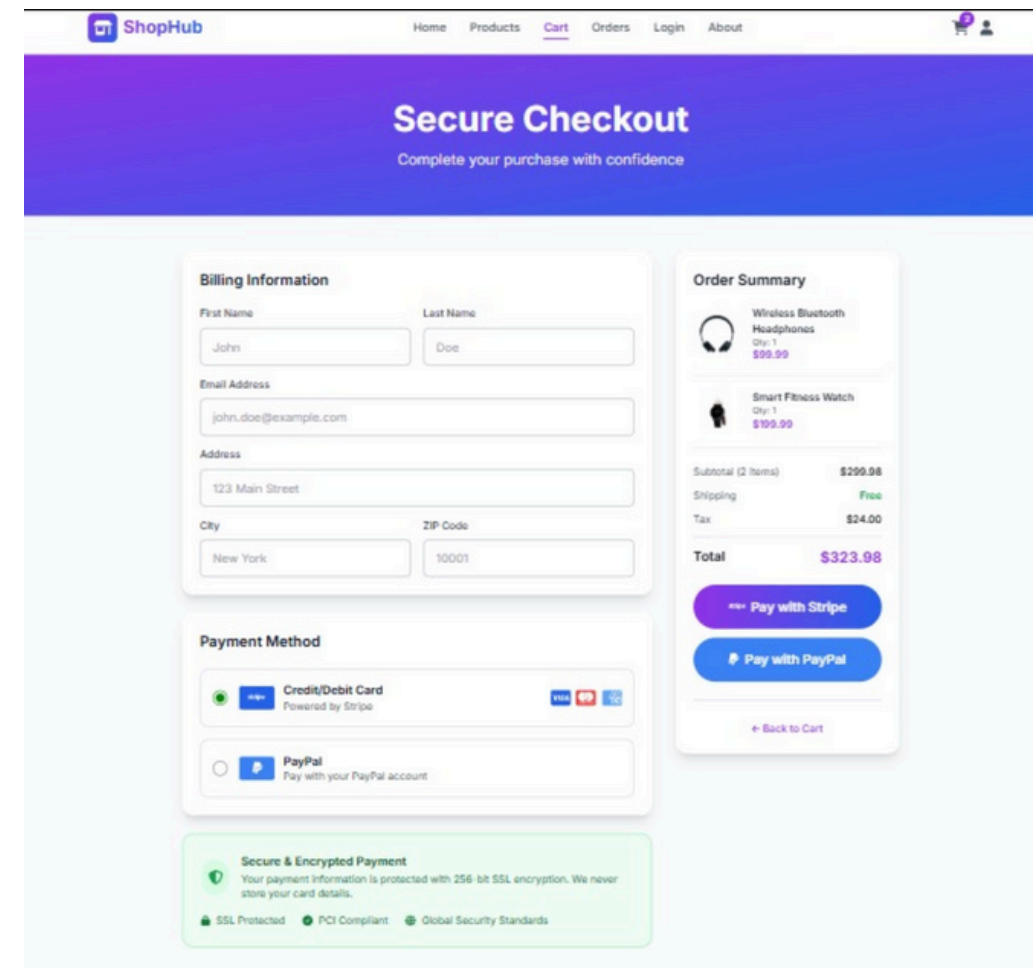
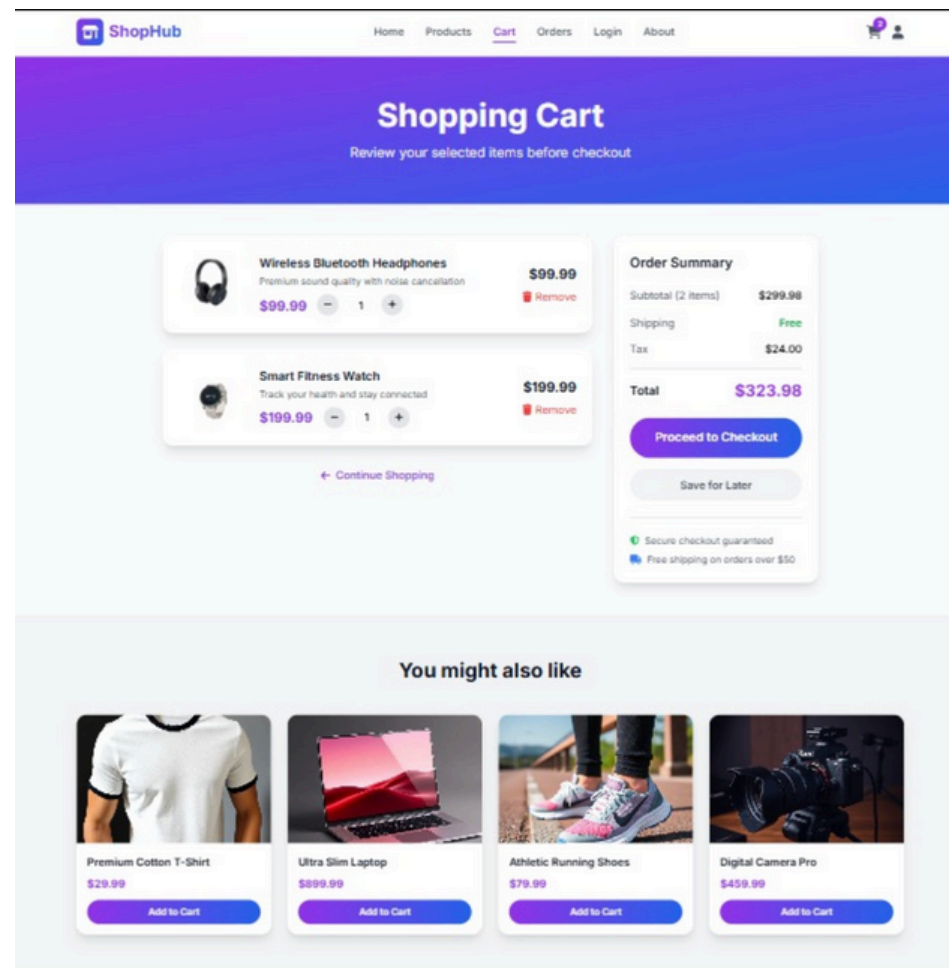
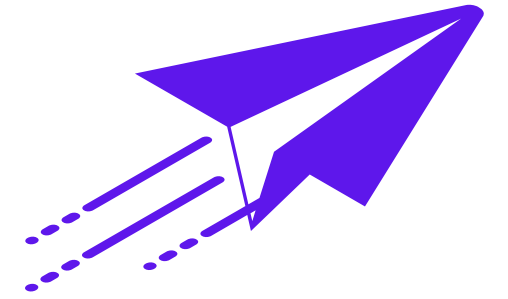
- Users can create an account or log in securely
- Authentication is required to place orders and view order history



## Products Page

- Displays a list of available products with images, prices, and ratings
- Users can filter products by category and sort them by relevance
- Each product card provides quick access to essential information
- “Add to Cart” action allows users to select products instantly

# USER EXPERIENCE: SHOPPING & ORDER MANAGEMENT



## Selection & Cart

- Users add products to the cart and manage quantities

## Checkout & Orders

- Users enter billing information
- Select a payment method (Card or PayPal – simulated)

## Order History

- Users can view previous orders
- Order status is displayed

# CONCLUSION & FUTURE IMPROVEMENTS

## Project Outcomes

- Successfully developed an online shopping system using the Waterfall methodology
- Implemented core features: authentication, product browsing, cart, checkout, and orders
- Applied MVC architecture with Factory and Strategy design patterns
- All critical test cases passed successfully

## Future Improvements

- Integration of real payment gateways
- Database persistence with Spring Data JPA
- Enhanced security and role-based access
- Advanced search, filtering, and order tracking

